County Housing Needs Forecasting Model v0.1: Design, Data and Build Plan

Oct 6, 2026 · @Gideon

Start here

Start by running the delivered code on its synthetic test data, then make the five decisions below, freeze this document as your v0.1 methodology, and only then download real data. The code, configuration and tests already exist; your first two weeks are setup, decisions and the first real data pull.

First two weeks, in order

1. Confirm with your immigration attorney, and against your employer's outside-work and IP policy, that you build this on your own time and equipment under a personal GitHub account.
2. Get a free Census API key. A HUD User token is optional (only for the CHAS API route).
3. Create a public GitHub repository and push the delivered code unchanged. That first commit is dated evidence of where the work started.
4. Run the synthetic test (README, Quick start). It proves Python, R and every module work on your machine, and that the two engines agree.
5. Make the five decisions below in `config/model.yaml` and `config/runs.yaml`.
6. Export this document to PDF as `methodology/methodology_v0.1.pdf` and commit it **before** running back-test B5 on real data. The commit date shows B5 tested a method fixed in advance.
7. Run `run_acquire.py` and `run_prepare.py`. Prepare stops with a message if any downloaded file does not match the expected layout; fix the config entry it names.

Decisions only you can make

Decision	Default	Where	Why it matters
Healthy vacancy by tenure	Owner 2.0%, renter 6.0% (low 1.5/5, high 2.5/8)	`model.yaml` vacancy	Normative; moves the existing gap directly

Decision	Default	Where	Why it matters
Count "other vacant" units as supply?	No (share 0)	`model.yaml` vacancy	Large in rural and shrinking counties
Weight of the two migration windows	50% 2015-2020, 50% 2020-2025	`runs.yaml` production	2020-2025 holds the pandemic moves and the 2022-2024 immigration surge
Headship range for low/high	Min/max of ACS 2020-2024, Census 2020, Census 2010	`runs.yaml` households	Sets the household-formation uncertainty
Small-county shrinkage K	100 people	`model.yaml` migration	Chosen in back-test A10 from 1,000, 250, 100, 25 and 0; never tuned on B5

What Version 0.1 produces

For every county with complete data, the model launches from the July 1, 2025 estimates and reports July 1, 2030 (5-year) and July 1, 2035 (10-year) values under low, base and high scenarios. Population and households also carry 80% prediction intervals calibrated on the back-tests.

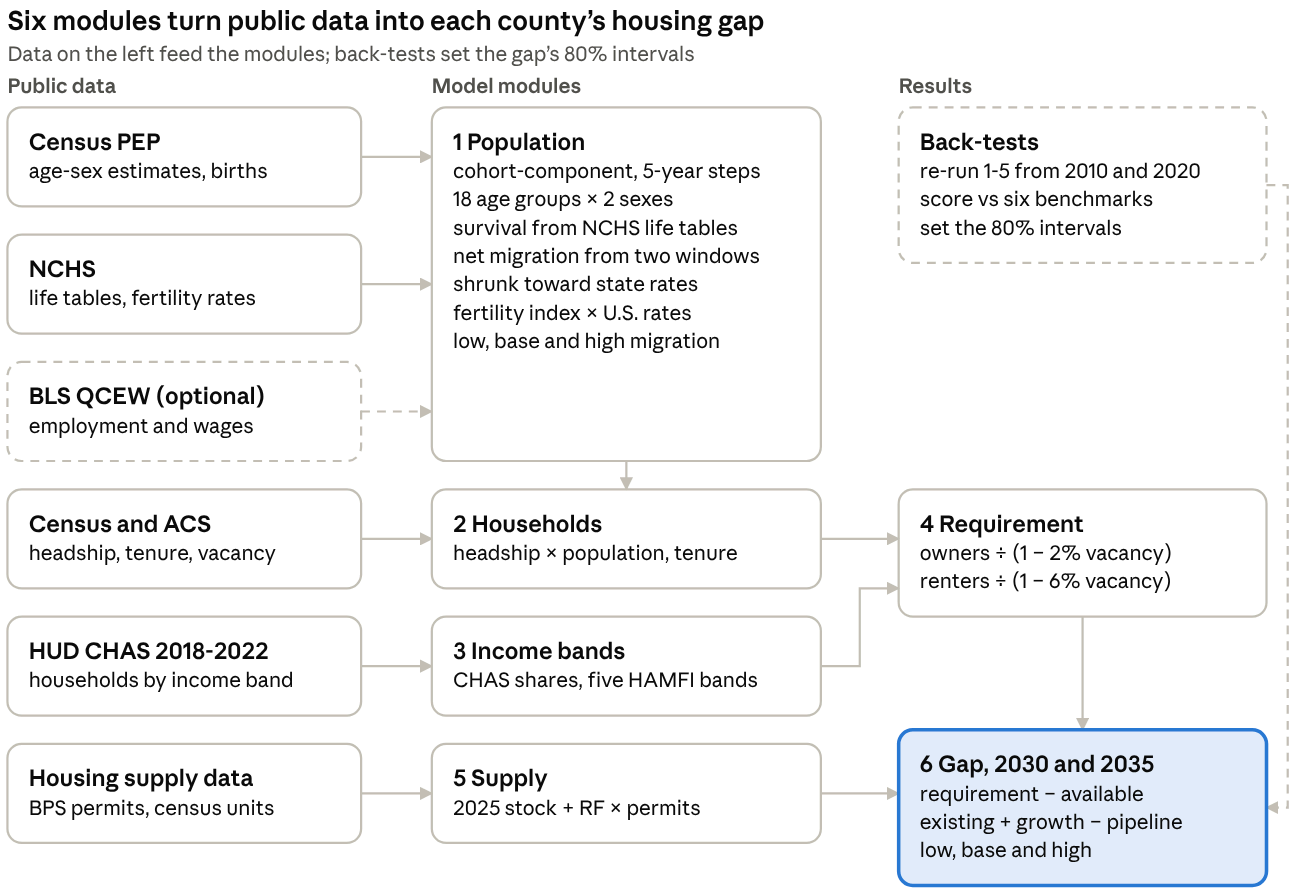
Output	Detail	Uncertainty shown
Population	Total; by sex and 5-year age group	Low/base/high migration; 80% interval
Households	Total, owner, renter	Scenarios; 80% interval (10-year)
Households by HUD income band	⩽30, 30-50, 50-80, 80-100, >100% HAMFI, by tenure; >80% combined	Scenarios
Housing-unit requirement	Total and by income band, with a vacancy allowance by tenure	Scenarios

Output	Detail	Uncertainty shown
Expected supply	Stock = 2025 units + realization factor × permits; year-round available units	Low/high permit volume
Gap	Requirement minus available units, split into existing gap, growth in need and pipeline	Composite scenarios + one-at-a-time sensitivity
Diagnostics	Every estimated parameter, raw and after shrinkage	—

Version 0.1 does not forecast affordable supply by income band. The band figures are households and the units they require, not a band-level gap; that is a Version 0.2 question.

Model architecture

Six public data sources feed five modules that end in the gap; the back-tests re-run the same modules from 2010 and 2020 launches and set the gap's intervals.



model architecture · 6 data sources, 6 modules

Employment (dashed) changes the migration level only if back-test B5 shows it helps; every other arrow is always on.

Population module

Population is projected by the cohort-component method in five-year steps for 18 age groups (0-4 to 85+) and two sexes. Net migration is learned from two observed windows and shrunk toward state rates where counties are small. Code: `mortality.py`, `migration.py`, `fertility.py`, `projection.py`.

Survival comes from NCHS single-year period life tables (person-years L, totals T, radix l0 = 100,000):

$$S_a = \frac{{}_5L_{5a+5}}{{}_5L_{5a}}\ (a=0,\dots,15), \qquad S_{16} = \frac{{}_5L_{85}}{{}_5L_{80}}, \qquad S_{17} = \frac{T_{90}}{T_{85}}, \qquad S_b = \frac{{}_5L_0}{5\,l_0}$$

Production uses the 2022 state tables; back-test A uses the U.S. 2010 table.

Net migration is the cohort residual over a window from t to t+n, expressed as a five-year rate on the starting cohort:

$$NM_{a+1} = P_{a+1}(t+n) - P_a(t)\,S_a, \qquad m_a = \frac{5}{n}\cdot\frac{NM_{a+1}}{P_a(t)}$$

The 85+ group pools ages 80-84 and 85+ as exposure. Where county births for the window are known, children born in it get their own rate from P(0-4) minus births × S_b; otherwise their rate is 0.5 × the 0-4 rate, since they are exposed for about half the period.

Small-county shrinkage blends each county's age-sex rate with its state's rate, weighted by the cell's population:

$$\tilde m_{c,a,s} = w\,m_{c,a,s} + (1-w)\,m_{\mathrm{state},a,s}, \qquad w = \frac{P_{c,a,s}(t)}{P_{c,a,s}(t) + K}$$

With K = 100, a cell of 100 people gets half its own rate and a cell of 900 gets 90%. K was chosen in back-test A10 from 1,000, 250, 100, 25 and 0: K = 100 tied for the lowest error and had less upward bias in small counties, which a stronger pull toward the state average (dominated by large, growing counties) produced.

Migration scenarios. Base averages the two window schedules (50/50). Low and high reproduce, county by county, the observed window that implies fewer or more net migrants. They are real, observed regimes, not arbitrary percentages.

Fertility uses national age-specific rates f scaled by a county index k (indirect standardization), shrunk toward the state with weight E / (E + 250):

$$k_c = \frac{B_c / E_c}{B_{US} / E_{US}}, \qquad E_c = \sum_{y}\sum_{a} f_a\,W_{c,a,y}, \qquad B_{t,t+5} = 5\,k_c\sum_{a} f_a\,\frac{W_a(t)+W_a(t+5)}{2}$$

B are births from the PEP components of change and W women aged 10-49. Fertility barely moves a 10-year housing forecast: children born after 2025 do not head households by 2035.

One projection step, with σ = 1.05/2.05 for boys and 1/2.05 for girls, and S + m floored at 0:

$$P_{a+1}(t+5) = P_a(t)\,(S_a + m_a), \qquad P_{0\text{-}4}(t+5) = B\,\sigma_s\,(S_b + m_b)$$

Households, tenure and income bands

Households come from age-specific headship and ownership rates applied to the projected population; income bands come from each county's CHAS distribution within tenure. Code: `households.py`, `income.py`.

For the nine householder age groups g (15-24, 25-34, 35-44, 45-54, 55-59, 60-64, 65-74, 75-84, 85+) used by ACS table B25007 and census tables H17 (2010) and H13 (2020):

```
HH_g(t) = HR_g\,P_g(t), \qquad Own_g(t) = OWN_g\,HH_g(t), \qquad Rent_g(t) = HH_g(t) - Own_g(t)
```

HR is householders ÷ population of that age and OWN is owner householders ÷ householders, both from the 2020-2024 ACS (B25007 ÷ B01001).

ACS rates are shrunk toward the state rate using their published margins of error (SE = MOE / 1.645). The between-county variance τ^2 is estimated by the method of moments for each age group, pooled nationally. Census-based rates have no sampling error and are not shrunk.

```
\widehat{HR}_c = w_c\,HR_c + (1-w_c)\,HR_{\mathrm{state}}, \qquad w_c = \frac{\tau^2}{\tau^2 + SE_c^2}
```

Headship scenarios. Low and high take, for each age group, the lowest and highest rate among the 2020-2024 ACS, the 2020 Census and the 2010 Census. They phase in linearly by 2035, so every scenario starts from the same 2025 households.

Income bands. Each tenure total is split by the county's CHAS 2018-2022 shares (Table 8), held constant in Version 0.1:

```
HH_{c,\tau,b}(t) = HH_{c,\tau}(t)\cdot\frac{CHAS_{c,\tau,b}}{\sum_{b'} CHAS_{c,\tau,b'}}
```

Known limitation: rate denominators include group-quarters residents (dorms, prisons, nursing homes). That flattens young-adult headship in college counties such as Dane, Whitman and Marquette; Version 0.2 removes group quarters from the denominator.

Supply, requirement and gap

The gap is the year-round market housing that projected households need, minus the units the county has plus what its permit pipeline is expected to add. It is reported in three parts so a reader can see whether a shortfall already exists or is coming. Code: `supply.py`, `gap.py`.

Requirement, with a normative vacancy allowance by tenure (default owner 2.0%, renter 6.0%):

$$Req(t) = \frac{Own(t)}{1 - v_{own}} + \frac{Rent(t)}{1 - v_{rent}}$$

Realization factor: net units added per unit authorized, measured census to census. PEP housing-unit estimates are themselves built from permits, so using them here would be circular.

$$RF_c = \frac{HU_c(2020) - HU_c(2010)}{\sum_{y=2010}^{2019} Permits_{c,y}}, \qquad Add_c(h) = RF_c \times \overline{Permits}_{c,\,2023\text{-}2025} \times h$$

RF is shrunk toward the state factor with weight P / (P + 1,000), where P is permitted units; it is bounded to [−1, 3] and flagged outside [0.5, 1.5]. Values above 1 are legitimate: manufactured homes need no permit, and some areas lie outside permit-issuing places. Counties with no permits in the window follow their 2010-2020 census trend instead. Low and high supply sustain the lowest and highest single year of 2023-2025 permits. The pipeline uses the latest three permit years because permitted units are usually built within a year or two; in back-test A10 a five-year window reached back into the housing boom and raised housing-unit error from 6.7% to 7.8%.

Available units remove stock that is not on the year-round market, using the county's ACS vacancy shares (α = share of "other vacant" counted as available, default 0):

$$Avail(t) = \left[HU(2025) + Add(t)\right]\left(1 - \frac{Seasonal + Migrant + (1-\alpha)\,OtherVacant}{HU}\right)$$

Without this step a seasonal-home county such as Vilas would show a large false surplus.

Gap and its exact decomposition (positive = shortfall):

$$Gap(t) = \underbrace{Req(2025) - Avail(2025)}_{\text{existing gap}} + \underbrace{Req(t) - Req(2025)}_{\text{growth in need}} - \underbrace{Avail(t) - Avail(2025)}_{\text{pipeline}}$$

The headline range is a composite: low = low migration, low headship, low vacancy target, high supply; high = the opposite. `sensitivity.csv` shows how far each assumption alone moves the 2035 gap.

Employment add-on

Employment enters only if it earns its place: it shifts the migration level in production only when it lowers out-of-sample error in back-test B5. The switch is `employment.use_in_production` in `model.yaml`, false until B5 says otherwise. Code: `employment.py`.

Estimated across all counties with state fixed effects and robust (HC1) standard errors:

```
mig_c = \alpha + \beta_1\,g^{emp}_c + \beta_2\,g^{wage}_c +
\beta_3\,mig^{prior}_c + \gamma_{state} + \varepsilon_c
```

mig is the five-year crude net migration rate of the target window; g are annualized log growth rates of QCEW total covered employment and average weekly wage over the years before that window.

Run	Fit on	Applied with	Scored against
Back-test B5	2015-2020 migration; 2010-2015 growth; 2010-2015 prior migration	2015-2019 growth; 2015-2020 prior migration	Vintage 2025, July 2025
Production	2020-2025 migration; 2015-2019 growth; 2015-2020 prior migration	2019-2025 growth; 2020-2025 prior migration	Future

The predicted crude rate replaces the base level: the base schedule is shifted by the difference, spread across ages in proportion to the state's absolute rates so it lands where migration happens. B5 reports the plain model and this `model_econ` variant side by side. If employment does not help, dropping it is itself a reportable finding.

Data you need

Fifteen public sources from four agencies, all free; `run_acquire.py` downloads every one into `data/raw/` and logs each file's URL, time, size and SHA-256. Every URL and code lives in `config/sources.yaml`; every processed variable is documented in `data/data_dictionary.csv`.

Component	Exact file or table	Vintage	Feeds	Source
Base population by age and sex	CC-EST2025-ALLDATA, one CSV per state: TOT_MALE, TOT_FEMALE by AGEGRP 1-18; YEAR 1 = Apr 1, 2020 base, 2-7 = Jul 1, 2020-2025	Vintage 2025	Launch population; 2020-2025 migration window	Census
Births	CO-EST2025-ALLDATA: BIRTHS2021-BIRTHS2025 (July-June years)	Vintage 2025	Fertility index; births term of migration	Census
2010-2020 population by age and sex	CC-EST2020INT-ALLDATA, one CSV per state	Intercensal	2010-2015 and 2015-2020 windows; tests A5, B5	Census
2000-2010 population by age and sex	CO-EST00INT-AGESEX-5YR: POPESTIMATE2000-2010, CENSUS2010POP	Intercensal	Test A launch and windows	Census
Births 2000-2020	CO-EST2009-ALLDATA and CO-EST2020-ALLDATA: BIRTHS columns	Vintages 2009, 2020	Fertility index in back-tests	Census
Census 2010 counts	SF1 API: P12, H3, H4, H5, H17	2010	Test A headship, vacancy, housing units	Census API
Census 2020 counts	DHC API: P12, H3, H4, H5, H13	2020	Test A targets; headship range; realization factor	Census API

Component	Exact file or table	Vintage	Feeds	Source
Census 2000 housing units	SF1 API: H001001	2000	Test A realization factor	Census API
Headship, tenure, vacancy	ACS 5-year API: B01001, B25007, B25001, B25003, B25004, estimates and margins of error	2020-2024	Production headship, ownership, vacancy shares	Census API
Housing-unit stock	CO-EST2025-HU.xlsx (Excel table; names matched to FIPS)	Vintage 2025	2025 base stock	Census
Permits	Building Permits Survey county annual files co{YYYY}a.txt, 2000-2025: units in 1, 2, 3-4 and 5+ unit structures	Annual	Realization factor; pipeline	Census
HUD income bands	CHAS county (050) CSV zip, Table 8	2018-2022, released Dec 23, 2025	Income-band shares	HUD User
Mortality	Life tables U.S. 2010 (NVSR 63-7), U.S. 2024 (NVSR 75-5), states 2019 (NVSR 70-18), states 2022 (NVSR 74-12); table 2 male, 3 female	2010-2024	Survival ratios	NCHS
National fertility rates	Births: Final Data for 2010, 2019 and 2024 (NVSR 61-1, 70-2, 75-2), typed into `config/national_asfr.csv` with citations	2009-2024	Projected births	NVSR 75-2
Employment and wages	QCEW annual averages, county total covered (agglvl 70): emplvl, weekly wage	2010, 2015, 2019, 2025	Employment add-on	BLS

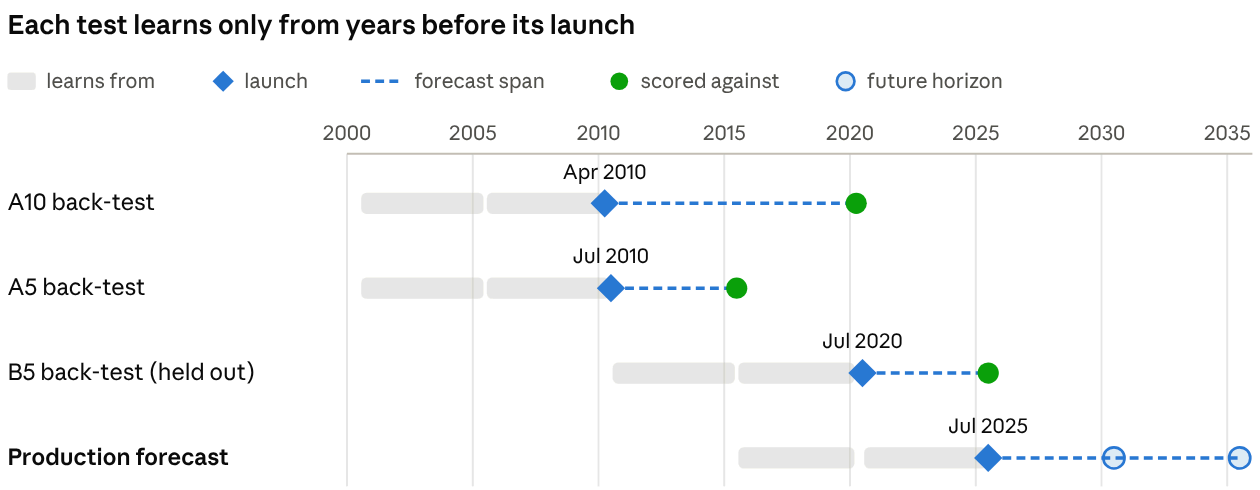
Check on your first download. My workspace could not reach these sites directly, so the downloaders follow the published layouts and were tested on fixture files built to the same layouts. Three items are marked VERIFY in `sources.yaml`, and the code checks each one for you:

- **Intercensal YEAR codes.** `run_prepare.py` compares the presumed April 2010 and April 2020 codes with the census counts and stops if they disagree.
- **CHAS Table 8 column map.** Prepare checks that the five bands sum to each tenure total and stops if they do not; confirm against the dictionary inside the zip.
- **File names that may differ.** These are the state life tables on the NCHS server and the CHAS zip URL. Acquire tries the common alternatives, then prints the landing page to download from by hand.

Validation design

Three back-tests run the full model from past launch dates, using only information dated at or before each launch, and score it on all counties (about 3,100; Connecticut and Puerto Rico excluded) against later census counts or estimates and six naive benchmarks. Code: `run_backtest.py`, `benchmarks.py`, `validation.py`.

Test	Launch	Learns from	Scored against	Measures	Role
A10	Apr 1, 2010 (Census 2010)	Migration 2000-2005 and 2005-Apr 2010; 2010 Census headship and vacancy; realization factor 2000-2010; permits 2007-2009; U.S. 2010 life table; 2009 fertility	Census 2020	Population, households, owners, renters, housing units	Main test; the only place K may be tuned; calibrates 10-year intervals
A5	Jul 1, 2010	Same windows	Intercensal Jul 1, 2015	Population	Calibrates 5-year intervals
B5	Jul 1, 2020	Migration 2010-2015 and 2015-2020; 2019 state life tables; 2019 fertility; permits 2017-2019	Vintage 2025, Jul 1, 2025	Population; housing units as a consistency check only	Held out: run once, after the method is frozen; tests interval coverage and the employment add-on



back-test and production timeline · 2000-2035

No grey calibration window runs past its own row's launch, and only the green dates are ever used to score the model.

Information-set rule (enforced by `tests/python/test_pipeline.py`):

- Every calibration window ends at or before launch. A10's second window stops at the April 2010 census count, not July 2010.
- Fertility rates come from the calendar year before launch (2009 rates for the 2010 launches).
- A10 also reports a diagnostic using the actual 2010-2019 permits, clearly labelled, never as the model's score.
- B5 starts from the Vintage 2025 estimate of July 2020, so forecast error is not mixed with revisions to the 2020 estimate. A strict real-time test would start from Vintage 2020; say so in the paper.

Benchmarks. Constant population; linear trend (LIN); compound growth over 10 years (EXP) and 5 years (EXP_RECENT); constant share of a linearly projected state (SHR); and the average of LIN, EXP and SHR (AVE). Households are also compared with a constant household-to-population ratio, and housing units with constant and linear trends.

Metrics, for all counties and by size (<10k, 10k-50k, 50k-250k, 250k+) and prior growth (declining, stable, growing): MAE, RMSE, MAPE, median APE, MALPE (bias), share within 5%, and the share of counties where the model beats each benchmark.

```
MAPE = \frac{100}{n}\sum\left|\frac{F-A}{A}\right|, \qquad MALPE = \frac{100}{n}\sum\frac{F-A}{A}, \qquad RMSE = \sqrt{\frac{1}{n}\sum (F-A)^2}
```

Prediction intervals are empirical. The 10th and 90th percentiles of percent error (q10, q90), by size class, from an earlier back-test of the same horizon give the 80% interval for a new forecast F: [F / (1 + q90/100), F / (1 + q10/100)]. Calibrated on A5, their coverage is checked on B5; about 80% of actuals should fall inside.

What to expect. Simple trend averages are often hard to beat on county total population. The model's value is what benchmarks cannot produce: age structure, households, tenure and income bands. Report the comparison honestly either way.

Development tuning (October 2026). Two settings were chosen on A10 and checked on A5 before B5 was run, and B5 informed no choice. Migration shrinkage K = 100 (from 1,000) lowered 10-year population error from 7.0% to 6.8% and bias from +4.8% to +4.0%; 5-year error fell from 3.9% to 3.8%. Using the latest three permit years instead of five lowered 10-year housing-unit error from 7.8% to 6.7%. The remaining upward bias reflects slower growth in the 2010s than in the 2000s, which the trend benchmarks share.

Demonstration counties

Sixteen counties in six states, chosen to span size, growth, age structure, seasonal housing and college towns. Neither Wisconsin county is in the seven-county SEWRPC region. The model runs nationally; these are a filtered view in `output/production/demo_counties.csv`, listed in `config/counties_demo.csv`.

FIPS	County	State	Why it is in the sample
55025	Dane	Wisconsin	Large, growing university county; data you know well
55125	Vilas	Wisconsin	Small seasonal-home county; tests the available-stock adjustment
04013	Maricopa	Arizona	Very large, fast-growing metro core
04019	Pima	Arizona	Large metro, slower growth, university
04025	Yavapai	Arizona	Retirement destination with an old age structure
26163	Wayne	Michigan	Very large county with long-run population and housing loss
26081	Kent	Michigan	Large, steadily growing metro

FIPS	County	State	Why it is in the sample
26103	Marquette	Michigan	Small, stable Upper Peninsula county with a university
37183	Wake	North Carolina	Large, very fast-growing metro
37119	Mecklenburg	North Carolina	Metro core with strong in-migration
37083	Halifax	North Carolina	Small, declining rural county
48201	Harris	Texas	Very large, diverse core with heavy international migration
48453	Travis	Texas	Large, fast-growing core
48209	Hays	Texas	Medium, very fast-growing exurban county
53063	Spokane	Washington	Regional metro outside a coastal market
53075	Whitman	Washington	Small college county; tests group-quarters distortion of headship

The code

The repository runs end to end today: 19 Python tests and 4 R tests pass, and the independent Python and R engines agree to within 5×10^{-12} (relative) on every forecast value. A 3,000-county run takes about a minute.

```
# Prove it runs (no downloads): synthetic data, back-tests, forecast, R
cross-check, tests
python python/make_synthetic.py
python python/run_backtest.py --data data/synthetic --out output/synthetic
python python/run_forecast.py --data data/synthetic --out output/synthetic
Rscript R/crosscheck.R data/synthetic output/synthetic
python -m pytest tests/python -q && Rscript tests/R/test_engine.R

# Real run
export CENSUS_API_KEY=your_key
python python/run_acquire.py      # downloads + provenance log
python python/run_prepare.py      # processed tables + layout checks
python python/run_backtest.py     # A10, A5, B5, benchmarks, interval
calibration
python python/run_forecast.py     # 2030 and 2035, all counties
Rscript R/crosscheck.R            # R engine agrees?
```

Path	What it holds
`config/model.yaml`	Every assumption: shrinkage constants, vacancy targets, scenario definitions
`config/runs.yaml`	Launch, calibration windows and targets for production and each back-test
`config/sources.yaml`	Every URL, API table and layout code
`python/hnf/`	Python engine, one module per component, plus acquisition and preparation
`R/hnf_engine.R`	R engine, written separately from the same equations
`R/crosscheck.R`	Fails if the two engines differ anywhere
`tests/`	Unit tests, parser tests on real-layout fixtures, end-to-end tests
`output/<run>/`	Results, diagnostics, sensitivity, and a manifest with each input's SHA-256

What the tests check

- Survival ratios are exact under constant mortality; the migration residual recovers the rates used to project.
- The downloader builds every URL and logs provenance against a mocked network; parsers read fixture files built like each real download: PEP CSVs, the housing-unit Excel table, two-header permit files, the CHAS zip, NCHS life-table sheets, Census API extracts and QCEW slices.
- On synthetic data: no missing outputs, low ⩽ base ⩽ high gap in every county, an exact gap decomposition, identical launch households across scenarios, and the information-set rule.

The synthetic data in `data/synthetic/` exist only to test software: made-up states 90-92, counties named SYNTHETIC. Never report their numbers.

Changes from the original plan

The design keeps the plan's structure and principles; these ten changes fix points a demographer or housing analyst reviewing it would raise.

Original plan	Changed to	Why
Births from women at the start of the step	Average of start and end women; national rates × county fertility index	Start-only exposure misstates births when the female population is changing
Migration from the 2020-2025 residual alone	Two five-year windows (2015-2020, 2020-2025); low/high = the observed windows	2020-2025 holds pandemic moves and an immigration surge then drop; one window is one regime
One 5% vacancy allowance	Owner 2% and renter 6%; seasonal, migrant and other-vacant units removed from supply	Seasonal-home counties would show false surpluses; the two markets need different allowances
Realization factor from permits vs later stock growth	Measured census to census (2010-2020), with a trend fallback where no permits are issued	PEP housing-unit estimates are built from permits, so comparing them with permits is circular

Original plan	Changed to	Why
ACS headship rates used as published	Empirical-Bayes shrinkage with ACS margins of error; scenario rates phased in by 2035	Small-county ACS rates are noisy; current households should not differ by scenario
Low/base/high scenarios as the uncertainty measure	Scenarios plus empirical 80% intervals, coverage tested out of sample	Scenarios bracket assumptions; intervals show how wrong past forecasts actually were
Test A uses information "through 2010"	Windows end at the April 2010 launch; fertility from 2009	July 2010 data and calendar-2010 rates post-date an April 2010 launch
Test B "earlier period to 2025"	Launch July 2020, held out until the method is frozen and committed	A test you can tune against is not out of sample
Milwaukee and Waukesha in the demonstration	Dane and Vilas	Both are outside the SEWRPC region, supporting separation from your employer's work
Python only	Python plus an independent R engine and an automatic cross-check	Two implementations that agree are strong evidence the code matches the equations

Connecticut is excluded in Version 0.1 because its county geography changed in 2022 and the series no longer line up; a crosswalk is on the Version 0.2 list.

Limitations and Version 0.2

Version 0.1 is deliberately simple where the data or the evidence cannot yet support more; each limitation below is stated in the methodology rather than hidden.

Limitations of Version 0.1

- Headship denominators include group-quarters residents.
- Income-band shares are held at their 2018-2022 values; there is no band-level supply or affordability gap.
- The realization factor is reduced-form: it does not separate completions, demolitions and manufactured-home placements.

- Future permits continue at recent levels; announced projects are not modelled.
- Back-test A uses the U.S. 2010 life table rather than state tables; production uses 2022 state tables.
- Small counties are less accurate: in A10, counties under 10,000 people were over-projected by about 4% in population and 9% in housing units even after tuning, partly because many lost housing units to demolition and abandonment, which a permit-based method cannot see.
- Back-test B is scored against estimates, not counts; its housing-unit comparison is a consistency check only.
- Connecticut is excluded; three renamed or merged counties are recoded; counties whose series break are skipped and listed.
- 2020 Census age data carry disclosure-avoidance noise, small except in the smallest counties.

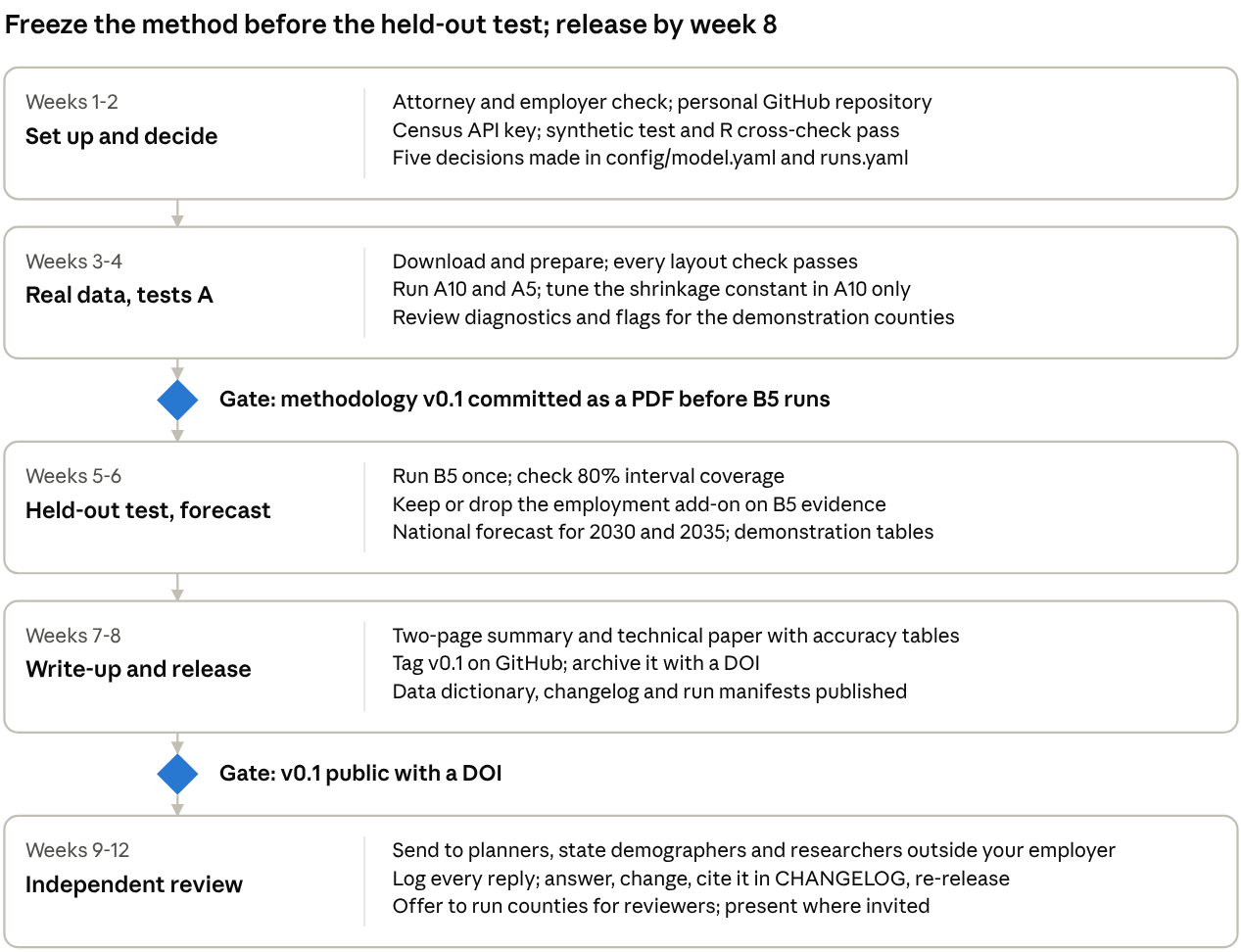
Version 0.2, in priority order

1. Remove group quarters from headship denominators (2020 Census group-quarters tables).
2. Add a Connecticut planning-region crosswalk.
3. Let income-band shares move with wage growth and household composition, tested by back-casting from CHAS 2008-2012 to 2018-2022.
4. Split supply into completions, losses and manufactured homes.
5. Shrink toward peer counties (size, growth, metro status) rather than the state: A10 showed that a strong pull toward the state average helps mid-size counties but biases small ones upward.
6. Swap in the 2021-2025 ACS when it is released.
7. Add an affordability gap by income band from the CHAS affordability tables.

Order items 3-7 by what reviewers ask for; each change cites its reason in `CHANGELOG.md`.

Twelve-week schedule and evidence log

Version 0.1 can be public with a DOI by week 8, leaving weeks 9-12 for outside review. The one date that must not slip out of order is the methodology commit before B5.



twelve-week plan · 5 phases, 2 gates

The two gates are what make the record credible: a frozen method before the held-out test, and a public, citable release before outreach.

Evidence to keep as you go

Evidence	Created	Where it lives
First commit of the repository	Week 1	GitHub history
`methodology_v0.1.pdf`, committed before B5	Week 4	`methodology/` and its commit date
Back-test outputs and run manifests (input SHA-256, config, version)	Weeks 4-6	`output/`, attached to the release
v0.1 release with DOI	Week 8	GitHub release and the archive record

Evidence	Created	Where it lives
Reviewer requests, replies and written comments	Weeks 9-12	A private evidence folder, not the public repository
Changes made in response, each citing its comment	As they happen	`CHANGELOG.md` and the commit that made the change
Requests to run a county, test a version or present	As they happen	The private evidence folder